SSP

Computer & Communication Department

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Modern Physics
Sheet 7
Solutions

Eng. Mohammad Alkammash

1. A hydrogen atom is in its ground state (n = 1). Using the Bohr theory of the atom, calculate (a) the radius of the orbit, (b) the linear momentum of the electron, (c) the angular momentum of the electron, (d) the kinetic energy, (e) the potential energy, and (f) the total energy.

Solution

Bohr radius (radius of the first level)	$a_0 = \frac{\hbar^2}{mke^2} = 5.29 \times 10^{-11}\, m = \ 0.529\, A$
Angular momentum	$L = n\hbar = 1.05 \times 10^{-34} kg.m^2/s$
Linear momentum	$P = \frac{L}{r} = 1.99 \times 10^{-24} kg.m/s$
Orbital speed	$v = \frac{P}{m} = 2.19 \times 10^{-6} m/s$
Kinetic energy	$K = \frac{1}{2} mv^2 = 2.18 \times 10^{-18} J = 13.6\, eV$
Total energy	$E_n = \frac{-13.6}{n^2}\ eV = -13.6\, eV$
Potential energy	$U = E - K = -27.3\, eV$

2. The electron in a hydrogen atom at rest makes a transition from the n = 2 energy state to the n = 1 ground state. Find the wavelength, frequency, and energy of the emitted photon.

Solution

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \ (\frac{1}{1^2} - \frac{1}{2^2})$$

$$\lambda = 1.215 \times 10^{-7} \ m = 121.5 \ nm$$

$$f = \frac{c}{\lambda} = 2.47 \times 10^{15} \ Hz$$

$$E = hf = 1.64 \times 10^{-18} \ J = 10.2 \ eV$$

3. The Balmer series for the hydrogen atom corresponds to electronic transitions that terminate in the state of quantum number n = 2. (a) Find the longest-wavelength photon emitted and determine its energy. (b) Find the shortest-wavelength photon emitted in the Balmer series.

Solution

- Any transition in the balmer series must have $n_f = 2$.
- Longest wavelength corresponds to minimum photon energy which corresponds to the smallest possible transition. ($n_i = 3$).

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\lambda = 6.56 \times 10^{-7}\, m = 656\, nm$$

$$E = \frac{hc}{\lambda} = 3.03 \times 10^{-19}\, J = 1.89\, eV$$

- Shortest wavelength corresponds to the maximum photon energy which corresponds to the largest possible transition. ($n_i = \infty$).

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \ (\frac{1}{2^2} - \frac{1}{\infty^2})$$

$$\lambda = 3.65 \times 10^{-7} \ m = 365 \ nm$$

4. (a) What value of n is associated with the Lyman series line in hydrogen whose wavelength is 102.6 nm? (b) Could this wavelength be associated with the Paschen or Brackett series?

Solution

Lyman series corresponds to $n_f = 1$

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$

$$\frac{1}{1.026 \times 10^{-7}} = 1.097 \times 10^7\ (\frac{1}{1^2} - \frac{1}{n^2})$$

$$n = 3$$

The Paschen series corresponds to $n_f = 3$.

Calculate the minimum wavelength of the Paschen series.

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$

$$\frac{1}{\lambda} = 1.097 \times 10^7\ (\frac{1}{3^2} - \frac{1}{\infty^2})$$

$$\lambda = 8.20 \times 10^{-7}\ m = 820\ nm$$

The minimum wavelength of the Paschen series is 820 nm.

So a photon wavelength 102.6 nm can't belong to Paschen or Brackett series.

5. A hydrogen atom initially in its ground state (n = 1) absorbs a photon and ends up in the state for which n = 3. (a) What is the energy of the absorbed photon? (b) If the atom returns to the ground state, what photon energies could the atom emit?

Solution

a) When a photon is absorbed, we replace the n_i and n_f in Rydberg formula.

$$\frac{1}{\lambda} = R(\frac{1}{n_i^2} - \frac{1}{n_f^2})$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \, (\frac{1}{1^2} - \frac{1}{3^2})$$

$$\lambda = 1.03 \times 10^{-7} \, m = 103 \, nm$$

$$E = \frac{hc}{\lambda} = 1.94 \times 10^{-18} \, J = 12.1 \, eV$$

b) The possible transitions are 3→1 or 3→2→1. So we have 3 different photons that can be emitted.

- 3→1: $E = 12.1 \, eV$
- 3→2: $E = 1.89 \, eV$
- 2→1: $E = 10.2 \, eV$

Note that $E_{3\rightarrow1} = E_{3\rightarrow2} + E_{2\rightarrow1}$

6. Calculate the frequency of revolution and the orbit of the electron in the Bohr model of hydrogen for n = 100, 1000, and 10000. (b) Calculate the photon frequency for transitions from the *n* to *n* - 1 states for the same values of *n* as in part (a) and compare with the revolution frequencies found in part (a). (c) Explain how your results verify the correspondence principle.

Solution

Frequency of revolution

$$f = \frac{v}{2\pi r} = \frac{P}{2\pi r m} = \frac{L}{2\pi r^2 m} = \frac{n\hbar}{2\pi r^2 m} = \frac{\hbar}{2\pi m a_0^2 n^3}$$

n = 100: $f = 6.59 \times 10^9\ Hz$
n = 1000: $f = 6.59 \times 10^6\ Hz$
n = 10000: $f = 6.59 \times 10^3\ Hz$

Frequency of transition

n = 100

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$
$$\frac{1}{\lambda} = 1.097 \times 10^7\ (\frac{1}{100^2} - \frac{1}{101^2})$$
$$\lambda = 0.0463\ m$$

$$f = \frac{c}{\lambda} = 6.48 \times 10^9\ Hz$$

n = 1000: $6.57 \times 10^6\ Hz$
n = 10000: $6.58 \times 10^3\ Hz$

As n → ∞, the quantum frequency of the transition approaches the classical frequency of revolution.